

NDX-4 (Q9U2M7) as a MutT-type 8-oxo-dGTP/8-oxo-GTP Sanitiser — Focused Curation Report

Gene: *ndx-4* (worm, *C. elegans*, NCBITaxon:6239) · **UniProt:** Q9U2M7 · **Structures:** PDB 1KT9, 1KTG **Focus:** function_assignment · **Hypothesis slug:** ndx-4-8oxo-dgtp-sanitiser

Executive Judgment

Partially supported — with the specific "MutT-type sanitiser" framing over-annotated.

The decisive structural question resolves **against** the seed hypothesis. NDX-4's substrate pocket is the **NUDT2-type adenine-stacking Ap4A-hydrolase pocket**, not a MutT/MTH1-type 8-oxoguanine-recognition pocket. This is supported by two independent public lines of evidence:

1. Structure (PDB 1KT9/1KTG, Bailey 2002,

P 11937063

the *C. elegans* Ap4A hydrolase binds its nucleotide substrate by **adenine ring-stacking** (explicitly noted in

P 20124691

and is tuned for dinucleoside **tetraphosphates**, not mononucleotide triphosphates. **Direct coordinate analysis of 1KTG performed here** shows the bound AMP adenine base is sandwiched by two tyrosines — **Tyr76** (3.36 Å, over N7) and **Tyr121** (3.34 Å, over C6) — plus His31 and Leu74, i.e. an aromatic ring-stacking pocket with no hydrogen-bond donor/acceptor arrangement poised to read the 8-oxoguanine O8/N7-H edge that MutT (Asn119) and human MTH1 rely on.

2. Sequence/evolution: NDX-4 is **49.0% identical to human NUDT2** (the canonical asymmetric Ap4A hydrolase) but only **25.0% / 20.3%** identical to *E. coli* MutT / human MTH1 — i.e. background-level for Nudix proteins that merely share the catalytic box. NDX-4 and NUDT2 share a near-identical substrate/Nudix-box segment (`WTPPKGHV_PGED...RET.EEA.I`).

The *in vitro* oxidized-nucleotide activity is real but **weak and non-specific**: ~2.5× preference for 8-oxo-dGTP over dGTP, and **no discrimination at all** for 8-oxo-GTP vs GTP (34.3% vs 33.7%), measured only in single-point assays at a very high enzyme:substrate ratio (1.51 μM enzyme / 20 μM substrate / 30 min), with **no kcat/Km reported** — versus a kinetically defined Ap4A hydrolase (Km 7 μM, kcat 27 s⁻¹). The reported 8-oxo-GTP → **8-oxo-GDP** product (loss of a single phosphate) is **not** the canonical MutT sanitisation chemistry (8-oxo-dGMP + PPi) and leaves a rephosphorylatable, incompletely "sanitised" product. The activity is also **redundant** (NDX-1, NDX-2, NDX-4 all show 8-oxo-GTPase/GDPase activity; NDX-2 does the canonical 8-oxo-dGDP→8-oxo-dGMP reaction;

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Net: NDX-4 genuinely possesses a moonlighting oxidized-guanine-nucleotide hydrolase side activity and contributes to genome stability *in vivo*

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but it is **not** a dedicated, structurally specialised MutT-type sanitiser. Its **primary/core function is the asymmetric Ap4A hydrolase**.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Co lim
P 11937063 (Bailey 2002)	Structural	Refutes MutT-pocket	Does the NDX-4 pocket recognise 8-oxoG?	Crystal structure (free + binary); pocket oriented for dinucleoside tetraphosphates "but not triphosphate" via precise adenine orientation	NDX-4 protein, PDB 1KT9/1KTG	Hi str of an an dG cor
P 20124691 (Jeyakanthan 2010)	Structural	Refutes MutT-pocket	Mode of nucleotide binding in this family	States C. elegans Ap4A hydrolase binds the adenine moiety by ring-stacking	Aquifex Ap4A hydrolase, references C. elegans	Hi bin mo spe inf
PDB 1KTG (this report, direct coord. analysis)	Structural/ computational	Refutes MutT-pocket	Which residues line the NDX-4 base pocket?	Bound AMP adenine base is sandwiched by Tyr76 (3.36 Å, over N7) and Tyr121 (3.34 Å, over C6) , with His31 and Leu74; aromatic ring- stacking, no O8- specific H-bond apparatus	NDX-4 binary complex, 1.80 Å	Hi ho ad (A 8-c cor (in ab rea
This report (public seq, NW alignment)	Computational / evolutionary	Refutes MutT-type assignment	Is NDX-4 a MutT/ MTH1 or NUDT2 ortholog?	49% id to NUDT2 vs 25%/20% to MutT/MTH1; shared NUDT2 Nudix-box segment	Q9U2M7 vs P50583/P08337/ P36639	Hi glo ali bu un
	Direct assay (kinetics)		What is the primary	Ap4A hydrolase, Km 7 µM, kcat 27	Recombinant C. elegans enzyme	Hi

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Con- lim
P 11738085 (Abdelghany 2001)		Supports core Ap4A function	catalytic activity?	s^{-1} , products AMP+ATP		
P 12475970 (Abdelghany 2003)	Direct assay / mutagenesis	Supports core Ap4A function	Catalytic residues	kcat 23 s^{-1} ; Nudix-box Glu residues essential; defines active site	Recombinant enzyme	Hi
P 21111690 (Arczewska 2011)	Direct assay + mutant phenotype	Supports (in vivo)	Does NDX-4 act as a MutT- type enzyme in vivo?	Hydrolyses 8-oxo- dGTP, suppresses E. coli mutT mutator , ndx-4 loss → genome instability + stress-gene upregulation	C. elegans + E. coli complementation	Me Hi viv gen ou act pro
P 24435662 (Sanada 2014)	Direct assay + mutant phenotype	Qualifies / competes	Is NDX-4 the dedicated sanitiser?	NDX-1, NDX-2, NDX-4 all have 8- oxo-GTPase/ GDPase activity; NDX-2 does canonical 8-oxo- dGDP→8-oxo- dGMP; checkpoint (chk-2/clk-2) response	C. elegans	Me sho rec & t is t Mu
P 12370170 (Fisher 2002)	Direct assay	Qualifies (promiscuity)	Substrate breadth of this Nudix	C. elegans Ap4A hydrolase active site is promiscuous (also a PRPP pyrophosphatase)	Recombinant enzyme	Me sho sim site hy sev sub

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Co
P 24347047 (silver NP study)	Review/indirect	Context only	—	Oxidative DNA damage-repair in <i>C. elegans</i> vs human; no direct NDX-4 active-site data	—	Lo rel act qu
P 28035004 (Waz 2017)	Structural (comparator)	Reference	What a true 8-oxoGTPase pocket looks like	MTH1 broad specificity via specific O8/N7 recognition of oxidized bases	Human MTH1	Hi the ND

GO Curation Implications (leads — require curator verification)

Verified current annotation state (QuickGO, Q9U2M7, this iteration): All existing MF/BP annotations are Ap4A-hydrolase-centric — **GO:0004081** *bis(5'-nucleosyl)-tetraphosphatase (asymmetrical) activity* (IDA,

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GO:0008796 (IEA/InterPro), GO:0016787 (IEA), plus BPs GO:0015967 (diadenosine polyphosphate catabolism), GO:0006167/0006172/0006754 (AMP/ADP/ATP biosynthesis), GO:0019693 & GO:0043135 (PRPP side activity), and GO:0006915 apoptosis (TAS). **Critically, there is NO existing 8-oxo-dGTP/8-oxo-GTP hydrolase MF or DNA-repair/sanitization BP**

annotation — so the seed hypothesis is a proposal to **add a currently-absent function**, not to modify an existing one. (Provenance: `ndx4_current_GO_annotations.csv`, `ndx4_GO_decision_table.csv`.)

- **Core MF — RETAIN as primary:** `bis(5'-nucleosyl)-tetrphosphatase (asymmetrical) activity` (GO:0004081; IEA parent GO:0008796). Strongly supported by kinetics

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and structure

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This remains the gene's primary molecular function and is already correctly annotated.

- **8-oxo-dGTP sanitiser MF — ADD ONLY AS SECONDARY / NON-CORE, qualified:** a term such as `8-oxo-7,8-dihydrodeoxyguanosine triphosphate pyrophosphatase activity` (GO:0035539) or an 8-oxo-GTP diphosphatase term is defensible from

P 21111690 /

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but should carry caveats: (a) in-vitro single-point evidence (no kcat/Km), (b) modest/absent base discrimination, (c) redundancy with NDX-1/NDX-2, and (d) a **product mismatch** — for 8-oxo-GTP the reported product is 8-oxo-GDP (single-phosphate release), which does not match a pyrophosphatase (PPi-releasing) definition. If only a single phosphate is released, GO:0035539 (a diphosphatase/pyrophosphatase term) may be the **wrong** term; a monophosphatase/"GDPase" framing would be needed. **Recommend curator confirm the product/chemistry before selecting the MF term.**

- **BP — ADD cautiously:** `DNA repair` (GO:0006281) / nucleotide-pool sanitisation / `GTP catabolic process`, supported *in vivo* by the mutT-complementation and genome-stability phenotype

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but flag as a secondary/context role given paralog redundancy.

- **Do NOT annotate** NDX-4 as a dedicated/orthologous **MutT/MTH1** enzyme by sequence orthology — it is a NUDT2 ortholog (49% id) and the MutT label reflects a functional moonlighting activity, not orthology.
- Avoid "protein binding" as an outcome; the informative MF is the Ap4A hydrolase activity.

Mechanistic Scope

Direct molecular activity: Mg^{2+} -dependent Nudix hydrolysis. The structurally and kinetically defined reaction is asymmetric cleavage of $Ap4A \rightarrow AMP + ATP$ (and related adenosine/dinucleoside polyphosphates with ≥ 4 phosphates; also PRPP). The oxidized-nucleotide hydrolysis uses the **same single active site**

([P 12370170](#))

shows this site is promiscuous), so 8-oxo-(d)GTP turnover is best interpreted as a low-specificity side reaction of the Ap4A pocket rather than an independent, optimised sanitiser site.

Downstream / inferred (not direct active-site evidence): genome stability, suppression of mutator phenotype, stress-responsive gene upregulation on *ndx-4* loss

([P 21111690](#))

and checkpoint-regulated post-embryonic development on oxidized-nucleotide accumulation

([P 24435662](#))

These are genetic/phenotypic outcomes consistent with — but not uniquely diagnostic of — a dedicated MutT-type active site.

Conflicts and Alternatives

- **Paralog redundancy / mis-attribution:** NDX-1, NDX-2 and NDX-4 share 8-oxo-nucleotide activity, and **NDX-2** carries out the canonical MutT product chemistry ($8\text{-oxo-dGDP} \rightarrow 8\text{-oxo-dGMP}$). The "dedicated worm MutT" role may belong more to NDX-2 than NDX-4

([P 24435662](#))

- **Orthology:** NDX-4 is a NUDT2 ortholog (adenine-stacking Ap4A hydrolase), not a MutT/MTH1 ortholog; the "MutT-type" descriptor is a functional analogy, not evolutionary identity.
- **Chemistry mismatch:** release of a single phosphate ($8\text{-oxo-GTP} \rightarrow 8\text{-oxo-GDP}$) differs from MutT/MTH1 pyrophosphate release; the product is incompletely sanitised.

- **Assay artefact risk:** single-point, high-enzyme assays can exaggerate weak activities; no k_{cat}/K_m or specificity constant has been reported for the oxidized substrates.

Knowledge Gaps

1. **Catalytic efficiency for 8-oxo-(d)GTP.** Checked: no k_{cat}/K_m in the cited literature. Matters: decides whether sanitisation is physiologically meaningful vs trace. Resolve with steady-state kinetics and a k_{cat}/K_m comparison to Ap4A and to NDX-2/MTH1.
2. **Reaction product identity (mono- vs diphosphate; PPi vs Pi).** Checked:

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 reports 8-oxo-GDP; dGTP product not clearly specified. Matters: determines the correct GO MF term. Resolve by HPLC/MS of products for 8-oxo-dGTP and 8-oxo-GTP.
3. **Direct structural proof of 8-oxoG recognition.** Checked: PDB 1KT9/1KTG are apo/Ap4A-analogue, not 8-oxo-dGTP complexes. Matters: the structural claim is currently inference from an adenine-stacking pocket. Resolve by co-crystal/docking of 8-oxo-dGTP, or mutagenesis of putative base-contact residues.
4. **In-vivo substrate.** Checked: genome-stability phenotype is genetic. Matters: distinguishes 8-oxo-dGTP sanitisation from Ap4A-signalling/stress roles. Resolve by measuring 8-oxo-dG/8-oxo-G in DNA/RNA and the nucleotide pool in *ndx-4* vs *ndx-2* mutants.

Discriminating Tests

- **Side-by-side steady-state kinetics** (k_{cat}/K_m) of NDX-4 for Ap4A, 8-oxo-dGTP, dGTP, 8-oxo-GTP, GTP, benchmarked against NDX-2 and human MTH1. A specificity constant for 8-oxo-dGTP within $\sim 10\times$ of MTH1 would support the hypothesis; $\geq 100\text{--}1000\times$ lower would refute a dedicated-sanitiser role.
- **Product analysis (LC-MS)** to confirm whether PPi or Pi is released (MutT-type vs NDP-forming).
- **Structure-guided mutagenesis / co-crystallography** with 8-oxo-dGTP to test for specific O8/N7-H base contacts (present in MutT/MTH1, predicted absent in NDX-4).

- **Genetic epistasis** of *ndx-4* vs *ndx-2* (single/double mutants) for 8-oxo-dG accumulation and mutation rate, to allocate the in-vivo sanitiser role among paralogs.

Curation Leads (require curator verification)

- **Retain** primary MF **GO:0004081** (asymmetric Ap4A hydrolase; IEA parent GO:0008796) — core function, already annotated IDA; refs

P 11738085

P 12475970

P 11937063

- **Note for curator:** the 8-oxo sanitiser function is currently **unannotated** in GO for *ndx-4*; this hypothesis would introduce it as a new term (see decision table below /

`ndx4_GO_decision_table.csv`).

- **Add, qualified/non-core**, a 8-oxo-guanine-nucleotide hydrolase MF — but **verify the product chemistry** before choosing between a pyrophosphatase term (e.g. GO:0035539) and a (di/mono)phosphatase term; refs

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- **Add, cautiously**, BP GO:0006281 (DNA repair) / nucleotide-pool sanitisation as a secondary role; ref

P 21111690

- **Flag** the "MutT-type" label as a functional analogy, not orthology (NDX-4 $\approx$ NUDT2 ortholog, 49% id).
- **Suggested curator questions:** (1) Is the 8-oxo-dGTP product 8-oxo-dGMP+PPi or 8-oxo-dGDP+Pi? (2) Is the dedicated worm MutT better assigned to NDX-2? (3) Should the 8-oxo activity be annotated with an IDA/in-vitro qualifier and a redundancy note?
- **Candidate references with snippets to verify:**

- P** 21111690

— "NDX-4 exhibits both hallmarks of a MutT-type enzyme with an ability to hydrolyze 8-oxodGTP and suppress the Escherichia coli mutT mutator phenotype."
- P** 24435662

— "NDX-1, NDX-2 and NDX-4 proteins have 8-oxo-GTPase or 8-oxo-GDPase activity."
- P** 11738085

— "It hydrolyses Ap4A with a K(m) of 7 microM and k(cat) of 27 s(-1)."
- P** 20124691

— "the adenine moiety of the nucleotide binds in a ring-stacking arrangement equivalent to that observed in the X-ray structure of Ap(4)A hydrolase from Caenorhabditis elegans."

Limitations

Analysis relied on public sequence, structure metadata, and abstracts; full-text kinetic tables and the atomic coordinates were not re-refined here. The structural argument against 8-oxoG recognition is strong but is inferred from binding mode + orthology rather than from a direct 8-oxo-dGTP·NDX-4 complex. Pairwise identities used a simple global alignment; values are approximate but the family assignment is unambiguous.
